

Practice Quiz: Cognition

Part A: Multiple Choice

1. Precision Π in a Gaussian distribution is: A) The mean of the distribution B) The inverse of the covariance matrix — $\Pi = \Sigma^{-1}$ C) The entropy of the distribution D) The mode of the distribution
2. The optimal precision π equals: A) *The prior precision* B) *The inverse of the expected squared prediction error* — $\pi = 1/E[(o - g(s))^2]$ C) Always 1 D) The number of observations
3. The Fisher information $I(\theta)$ for a Gaussian with known mean and unknown precision equals: A) The variance B) The precision — $I = \pi$ C) Zero D) The log-likelihood
4. Attention in the predictive coding framework is mathematically implemented as: A) Random amplification of signals B) Optimization of precision parameters — increasing gain on relevant prediction errors C) Suppression of all prediction errors D) Duplication of neural activity
5. When sensory precision $\pi_1 \gg$ prior precision π_2 , belief updates are: A) Driven primarily by priors B) Driven primarily by sensory data — the agent "trusts" its senses more C) Unaffected by either D) Driven by random noise
6. Log-precision $\gamma = \ln \pi$ is preferred over raw precision π because: A) It is always zero B) It ranges over all real numbers (avoiding the constraint $\pi > 0$) and provides more stable gradient dynamics C) It is easier to measure experimentally D) Neural systems cannot represent positive numbers
7. The Cramér-Rao bound states that: A) Precision is always bounded above B) The variance of any unbiased estimator is at least $1/I(\theta)$ — higher precision $\leftrightarrow$ lower minimum variance C) Fisher information is always negative D) The posterior is always Gaussian

Part B: Short Answer

1. For a Gamma prior $p(\pi) = \text{Gamma}(\alpha_0, \beta_0)$ on precision, with N Gaussian observations having known mean, derive the posterior parameters α_N and β_N . Show that the posterior mean precision increases with the number of observations.
2. In a two-level hierarchy, the precision ratio π_1/π_2 determines the balance between sensory and prior influence. Compute the posterior mean μ^* as a function of this ratio for a linear Gaussian model. What happens in the limits $\pi_1/\pi_2 \rightarrow \infty$ and $\pi_1/\pi_2 \rightarrow 0$?
3. Explain how a volatile environment (where precision changes over time) requires a hierarchical precision model. What additional level of inference is needed, and what does it correspond to neurally?